6/25/26

$^{12}/_{12}$ 100% Excellent!

Physics II: Exam 5
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These questions assess your understanding of the material from Chapters 23, 24 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. For questions with numerical calculations, you must show each major step necessary to find your final answer. For questions without numerical calculations, you must include clear and correct reasoning to support your answer. Remember to report the units and correct number of significant figures in your answers.

1. A generator coil is rotated 0.30 revolutions in 6.8 ms. The 260-turn circular coil has a 3.6-cm radius and is in a uniform 1.4-T magnetic field. What is the maximum emf that is induced?

ANSWER: 410.817 V ✓

$$\mathcal{E}(t)_0 = NAB\omega = N\pi r^2 B\omega$$

$$\frac{0.3\text{ rev}}{6.8\text{ ms}}\cdot\frac{2\pi\text{ rad}}{1\text{ rev}}\cdot\frac{1000\text{ ms}}{1\text{ s}} = 277.199\ \frac{\text{rad}}{\text{s}}$$

$$\mathcal{E}(t)_0 = (260)\pi(0.036\text{ m})^2\left(1.4\ \frac{N\cdot s}{C\cdot m}\right)\left(277.199\ \frac{\text{rad}}{s}\right) = \boxed{410.817\text{ V}}$$

2. A generator coil is rotated 0.10 revolutions in 9.8 ms. The 82.-turn circular coil has a 4.3-cm radius and is in a uniform 0.54-T magnetic field. What is the average emf that is induced?

ANSWER: 5.013 V ✓

$$\mathcal{E} = \frac{-N\Delta\Phi}{\Delta t} = \frac{-N(B_f A_f \cos\theta_f - B_i A_i \cos\theta_i)}{\Delta t}$$

$$\mathcal{E} = \frac{-NB\pi r^2(\cos\theta_f - \cos\theta_i)}{\Delta t} = \frac{-(82)(0.54\ \frac{N\cdot s}{C\cdot m})\pi(0.043\text{ m})^2(\cos(36^\circ) - \cos(0))}{0.0098\text{ s}} = \boxed{5.013\text{ V}}$$

3. Suppose a 1.7-H inductor is in series with a 5.4-Ω resistor and a current of 11. A flows through them. Find the current that flows through the inductor and resistor 3.8 s after the battery is disconnected.

ANSWER: 6.298×10^{-5} A ✓

$$I(t) = I_0 e^{\frac{-t}{\tau}} = I_0 e^{\frac{-t\cdot R}{L}} = (11\text{ A})e^{\frac{-(3.8\text{ s})(5.4\ \Omega)}{1.7\ \Omega\cdot s}} = \boxed{6.298\times10^{-5}\text{ A}}$$

+3

4. What is the maximum strength of the magnetic field in an electromagnetic wave that has a maximum electric-field strength of 1.1×10^3 V/m?

ANSWER: 3.667×10^{-6} T

$$c = \frac{E}{B} \rightarrow B = \frac{E}{c} = \frac{1.1 \times 10^{3} \frac{N}{C}}{3 \times 10^{8} \frac{m}{s}} = \boxed{3.667 \times 10^{-6} \text{ T}}$$

5. On its highest power setting, a 890-W microwave oven projects microwaves onto a 0.46-m-by-0.46-m area. Calculate the peak electric-field strength in the electromagnetic waves emitted by the microwave oven.

ANSWER: 1780 V/m

$$I_{ave} = \frac{c\varepsilon_0 E_0^2}{2} = \frac{P}{A} = \frac{P}{lw} \rightarrow E_0 = \sqrt{\frac{2P}{c\varepsilon_0 lw}} = \sqrt{\frac{2(890 \text{ A}\cdot\text{V})}{(3\times10^8 \frac{m}{s})(8.85\times10^{-12} \frac{C}{V\cdot m})(0.46 \text{ m})(0.46 \text{ m})}}$$

$$E_0 = \boxed{1780 \frac{N}{C}} \quad \frac{A\cdot V\cdot s\cdot V}{C\cdot m^2} = \frac{C\cdot N^2\cdot m^2\cdot s}{s\cdot C^2\cdot m^2\cdot C} = \frac{N^2}{C^2}$$

6. Calculate the motional emf induced along a 19.-km-long conducting object moving at an orbital speed of 3.4 $\frac{km}{s}$ perpendicular to Earth's 49.-μT magnetic field.

ANSWER: 3165.4 V

$$\mathcal{E} = Blv = (49\times10^{-6} \frac{N\cdot s}{C\cdot m})(19\times10^{3} \text{ m})(3.4\times10^{3} \frac{m}{s}) = \boxed{3165.4 \text{ V}}$$

7. How much energy is stored in a 1.3-mH inductor when a 59.-A current flows through it?

ANSWER: 2.26265 J

$$E_{ind} = \frac{1}{2}LI^2 = \frac{1}{2}(0.0013 \frac{V\cdot s}{A})(59 \text{ A})^2 = \boxed{2.26265 \text{ J}}$$

$$V\cdot s\cdot A = \frac{J\cdot \cancel{s}\cdot \cancel{C}}{\cancel{C}\cdot \cancel{s}} = J$$

+4

8. Suppose a 170-turn coil lies in the plane of the page in a uniform magnetic field that is directed into the page. The coil originally has an area of $0.44\,m^2$. It is then stretched for 0.20 s until it has no area. What is the magnitude of the induced emf if the uniform magnetic field has a strength of 0.082 T?

ANSWER: 30.668 V ✓

$$\mathcal{E} = \frac{-N\Delta\Phi}{\Delta t} = \frac{-N(B_f A_f \cos\theta_f - B_i A_i \cos\theta_i)}{\Delta t} = \frac{-NB\cos\theta(A_f - A_i)}{\Delta t}$$

$$\mathcal{E} = \frac{-(170)(0.082\ \frac{N\cdot s}{C\cdot m})\overset{=1}{\boxed{\cos(0)}}(0\ m^2 - 0.44\ m^2)}{0.2\ s} = \boxed{30.668\ V}$$

9. A portable x-ray unit has a step-up transformer, the 130 V-input of which is transformed to the 1.4×10^5-V output needed by the x-ray tube. The primary has 11. loops and draws a current of 8.4 A when in use. Find the current output of the secondary.

ANSWER: 0.0078 A ✓

$$\frac{V_s}{V_p} = \frac{I_p}{I_s} \rightarrow I_s = \frac{I_p V_p}{V_s} = \frac{(8.4\ A)(130\ \cancel{V})}{1.4\times10^5\ \cancel{V}} = \boxed{0.0078\ A}$$

10. Calculate the self-inductance of a 0.26 m-long, 0.044 m-diameter solenoid that has 140 coils.

ANSWER: 1.440×10^{-4} H ✓

$$L = \frac{\mu_0 N^2 A}{l} = \frac{\mu_0 N^2 \pi r^2}{l} = \frac{\mu_0 N^2 \pi d^2}{4l}$$

$$L = \frac{(\cancel{4}\pi^2\times10^{-7}\ \frac{N\cdot s\cdot m}{C\cdot m\cdot A})(140)^2\,\cancel{\pi}(0.044\ m)^2}{\cancel{4}(0.26\ m)} = \boxed{1.440\times10^{-4}\ H}$$

$$\frac{N\cdot s\cdot m}{C\cdot A} = \frac{V\cdot s}{A} = H$$

11. A battery charger meant for a series connection of several nickel-cadmium batteries needs to have a 16.-V output to charge the batteries. It uses a step-down transformer with a 240-loop primary and a 120-V input. If the charging current is 21. A, what is the input current?

ANSWER: 2.8 A ✓

$$\frac{V_s}{V_p} = \frac{I_p}{I_s} \rightarrow I_p = \frac{I_s V_s}{V_p} = \frac{(21\ A)(16\ \cancel{V})}{120\ \cancel{V}} = \boxed{2.8\ A}$$

+4

12. What is the characteristic time constant for a 3.2-mH inductor in series with a 0.83-Ω resistor?

ANSWER: 3.855×10^{-3} s

$\tau = \frac{L}{R} = \frac{0.0032\ \Omega \cdot s}{0.83\ \Omega} = \boxed{3.855 \times 10^{-3}\ s}$

✓

+1

Physics II: Exam 5

1. $\varepsilon = 411.\,\mathrm{V} \because \varepsilon = NAB\omega \sin(\omega t),\ \sin(\omega t) = 1$
2. $\varepsilon = 5.01\,\mathrm{V} \because \varepsilon = -N\frac{\Delta\Phi}{\Delta t}$
3. $I = 6.30 \times 10^{-5}\,\mathrm{A} \because I(t) = I_0 e^{-t/\tau},\ \tau = \frac{L}{R}$
4. $B_{max} = 3.67 \times 10^{-6}\,\mathrm{T} \because c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = \frac{E}{B} = f\lambda$
5. $E_0 = 1.78 \times 10^{3}\,\frac{\mathrm{V}}{\mathrm{m}} \because I_{ave} = \frac{c\varepsilon_0 E_0^2}{2} = \frac{cB_0^2}{2\mu_0} = \frac{E_0 B_0}{2\mu_0}$
6. $\varepsilon = 3.17 \times 10^{3}\,\mathrm{V} \because \varepsilon = B\ell v$
7. $E_{ind} = 2.26\,\mathrm{J} \because E_{ind} = \frac{1}{2}LI^2$
8. $\mathrm{emf} = 30.7\,\mathrm{V} \because \mathrm{emf} = -N\frac{\Delta\Phi}{\Delta t}$
9. $I_s = 7.80 \times 10^{-3}\,\mathrm{A} \because \frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}$
10. $L = 1.44 \times 10^{-4}\,\mathrm{H} \because L = N\frac{\Delta\Phi}{\Delta t} = \frac{\mu_0 N^2 A}{\ell}$
11. $I_p = 2.80\,\mathrm{A} \because \frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}$
12. $\tau = 3.86 \times 10^{-3}\,\mathrm{s} \because \tau = \frac{L}{R}$